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Aim

To study and execute commands for monitoring, Managing and terminating system process

Theory

ps: (Process states)

→ Reports a snapshot of current running processes and their Process IDs (PIDs).

→ Syntax: ps [options]

Top:

→ Provides a dynamic, real-time view of a running system, displaying system summary information and a list of processes currently being managed by the Linux kernel.

→ Syntax: top

htop:

→ An interactive, colorized processes viewer that allows vertical and horizontal scrolling to see all processes and complete command lines.

→ syntax: h top

kill:

→ sends a specific signal to a processes, most commonly used to gracefully or forcefully terminate it using it's PID.

→ Syntax: kill [signal] [PID].

→

nice:

→ used to launch a program with a modified scheduling

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priority (niceness value ranges from -20 to 19).
-> Syntax: nice -n [niceness value] [command]

-> bg & fg:
-> Used for job control. bg resumes a suspended job in the background, while fg brings a background job to the foreground.
-> Syntax: bg %[job_number] or fg %[job_number]

commands

$ sleep 1000 &
[1] 4512

$ ps
PID TTY TIME CMD
3201 pts/0 00:00:00 bash
4512 pts/0 00:00:00 sleep
4513 pts/0 00:00:00 ps

$ kill -9 4512
[1]+ killed sleep 1000

$ nice -n 10 top
(Top opens with lower priority. Press 'q' to quit)

$ sleep 500
(press Ctrl+Z to suspend the process)
[1]+ Stopped sleep 500

$ bg %1

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[1]+ sleep 500&

$ fg %1
sleep 500
(press ctrl +C to terminate)

Conclusion
System process monitoring and job control were successfully studied. Real time monitoring was observed (top), background and foreground job switching was performed (bg, fg), and specific background processes were successfully targeted and terminated using their Process IDs (ps, kill).

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Aim

TO Study and execute commands for scheduling automated tasks and managing user/group accounts.

Theory

1) cron / crontab

-> A time-based job scheduler used to run scripts or commands at specified intervals, dates or times.

-> Syntax: crontab -e (to edit the current user's cron jobs).

2) at

-> Used to schedule a command or script to be executed exactly once at a specific time in the future.

-> Syntax: at [time] (e.g., at 10:30PM).

3) useradd

-> Creates a new user account on the Linux system.

-> Syntax: sudo useradd [options] [username] (e.g., -m to create a home directory).

4) usermod

-> Modified a user account's properties, such as adding them to a new group.

-> Syntax: sudo usermod -aG [groupname] [username]

5) passwd

-> used to set or update a user's authentication token (password).

-> Syntax: sudo passwd [username]

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groupadd

→ Creates a new user group on the system.

→ Syntax: sudo groupadd [group name].

Commands

$ sudo groupadd developers
$ sudo useradd -m testuser1
$ sudo passwd testuser1
New password:
Retype ~~Pass~~ new password:
password: password updated successfully.
$ sudo usermod -aG developers testuser1
$ cat /etc/group | grep developers
developers: x: 1002; testuser1
$ at 05:00 PM
warning: commands will be executed using /bin/sh
at> echo "system update reminder" > /home/student/reminder.txt
at> <EOT> (Press Ctrl + D to save) job 1 at Wed Mar 19 17:00:00 2025
$ crontab -e [Inside the editor, add the following line to run a backup script everyday at 2 AM) 0 2 * * * /home/student/backup.sh
(save and exist) crontab: installing new crontab.

Conclusion

user and group administration tasks, including creating accounts, setting passwords, and assigning group memberships, were successfully executed.

Aim

TO study and execute commands for installing, updating, and removing software-package in Linux.

Theory

Linux package managers automate the process of downloading installing, upgrading and removing software along with it's required dependancies. Diff^n Linux distributions use diff^n package management systems.

apt (Advanced Package Tool)

The default package manager for Debian and Ubuntu based distributions.

Syntax:

sudo apt update (Updates the Local list of available packages and their versions).

sudo apt install [package_name] (Installs a new package).

Sudo apt remove [package_name] (Removes an installed package).

sudo apt upgrade (Installs newer versions of all currently installed packages).

yum/dnf

The package mangers used in Red Hat, CentOS, and Fedora-based systems (dnf is the modern successor to yum).

Syntax: Sudo dnf install [package_name]

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Snap

→ A universal, distribution-agnostic package management system developed by Canonical.

→ Syntax: sudo snap install [package_name]

Commands

$ sudo apt update
Hit:1 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease [119 kB]
Fetched 119 kB in 1s (101 kB/s)
Reading package lists... Done

$ sudo apt install tree
Reading package lists... Done
Building dependency tree... Done
The following New package will be installed:
tree
Do you want to continue? [Y/n] y
setting up tree (2.0.2-1)...
$ tree --version
tree v2.02 (c) 1996-2022 by steve Baker, Thomas Moore, Francesc Rocher, Florian Sesser, kyosuke Tokoro
$ sudo apt remove tree
Reading package lists... Done
Building dependency tree... Done
The following packages will be REMOVED:
tree

Do you want to continue? [Y/n] y
Removing tree (2.0.2-1)...

Conclusion

The fundamental processes of software administration were successfully executed. The local package repository list was refreshed, and a sample utility (tree) was successfully downloaded, installed verified and subsequently removed using the apt package manager.

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Aim

To study and execute commands for networking configuration, troubleshooting

Theory

ifconfig/ip

→ Displays or configure network interface parameters (like IP addresses and MAC addresses). ip is the modern networking command that replaces ifconfig.

→ Syntax: ifconfig or ip addr show.

Ping

→ Tests network connectivity btwn the local system and a remote host by sending ICMP ECHO_REQUEST packets.

→ Syntax: ping -c [count] [hostname_or_IP].

netstat/ss

→ Displays active network connections, routing tables, and listening ports.

→ Syntax: netstat -tuln (shows listening TCP/UDP ports).

wget/curl

→ command-line network downloaders used to fetch files from the internet using HTTP, HTTPS, or FTP protocols.

→ Syntax: wget [URL] or curl [URL]

ssh

→ A cryptographic network protocol used for secure remote login and command execution on another Linux machine.

→ Syntax: ssh [username]@[remote-ip-address]

Scp (secure copy)

→ securely transfers file btwn a local host and remote host over the SSH protocol.

→ Syntax: scp [source_file] [username]@[remote-ip]:[destination-path]

Commands

$ ip addr show

1: lo: <LOOPBACK, UP, LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN inet 127.0.0.1/8 scope host lo

2: eth0: <BROADCAST, MULTICAST, UP, LOWER_UP> mtu 1500 qdisc fq-codel state UP
inet 192.168.1.10/24 brd 192.168.1.255 scope global dynamic eth0

$ ping -c 3 google.com

PING google.com (142.250.193.46) 56(84) bytes of data.

64 bytes from bom12s18-in-f14.1e100.net: icmp_seq=1 ttl=116 time=14.2ms

64 bytes from bom12s18-in-f14.1e100.net: icmp_seq=2 ttl=116 time=15.1ms

64 bytes from bom12s18-in-f14.1e100.net: icmp_seq=3 ttl=116 time=14.8ms

$ netstat -tuln | head -n 4

Active Internet connections (only servers)

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Proto	Recv-Q	Send-Q	Local Address	Foreign address
tcp	0		0.0.00.0:22	0.0.0.0:* LISTEN

~~state~~

$ wget https://raw.githubusercontent.com/torvalds
/linux/master/README
2026-03-19 15:30:00 (15.5 MB/s) - 'README' saved
[1500/1500]

$ ssh student@192.168.1.50
Student@192.168.1.50's password:
Welcome to Ubuntu 22.04 LTS!
$ exit
logout

Conclusion

various network management and troubleshooting tools were practically explored. Local IP addresses were identified (ip addr), internet connectivity was verified (ping), active network ports were listed (netstat), a file was downloaded from the web (wget), and a secure remote shell session was establishment (ssh).

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